partition is then uniquely identified by the root of the tree associated with the partition. The advantage of trees over lists is that, if the tree is balanced, any operation on the tree is logarithmic (instead of linear, as for a list).

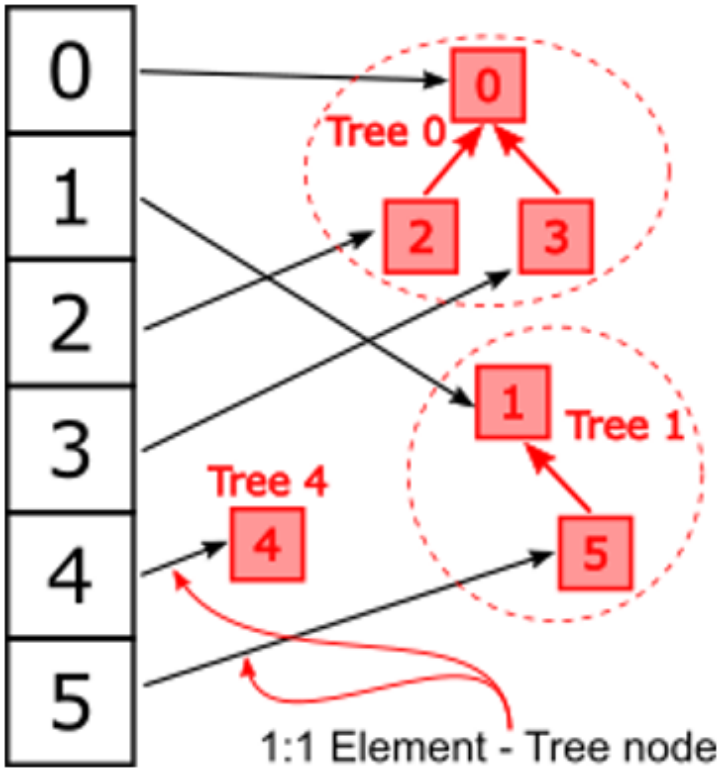


Figure 5.7 Representing a disjoint set with trees. Trees are named after their root, because we use the tree root as a unique identifier for the partition (we can do so under the assumption that elements are unique). Each element in the array points to a tree node: in the naïve implementation there is a1:1 mapping between elements and tree nodes. This means that, to get to the root of the tree, we might need to climb up the whole tree (and, on average, half of the height of the tree).